olist

NEWTON SCHOOL OF TECHNOLOGY | DVA CAPSTONE 2

# Olist Marketplace Analytics

Decoding the drivers of customer satisfaction and revenue

Sector: E-Commerce / Retail (Brazil)
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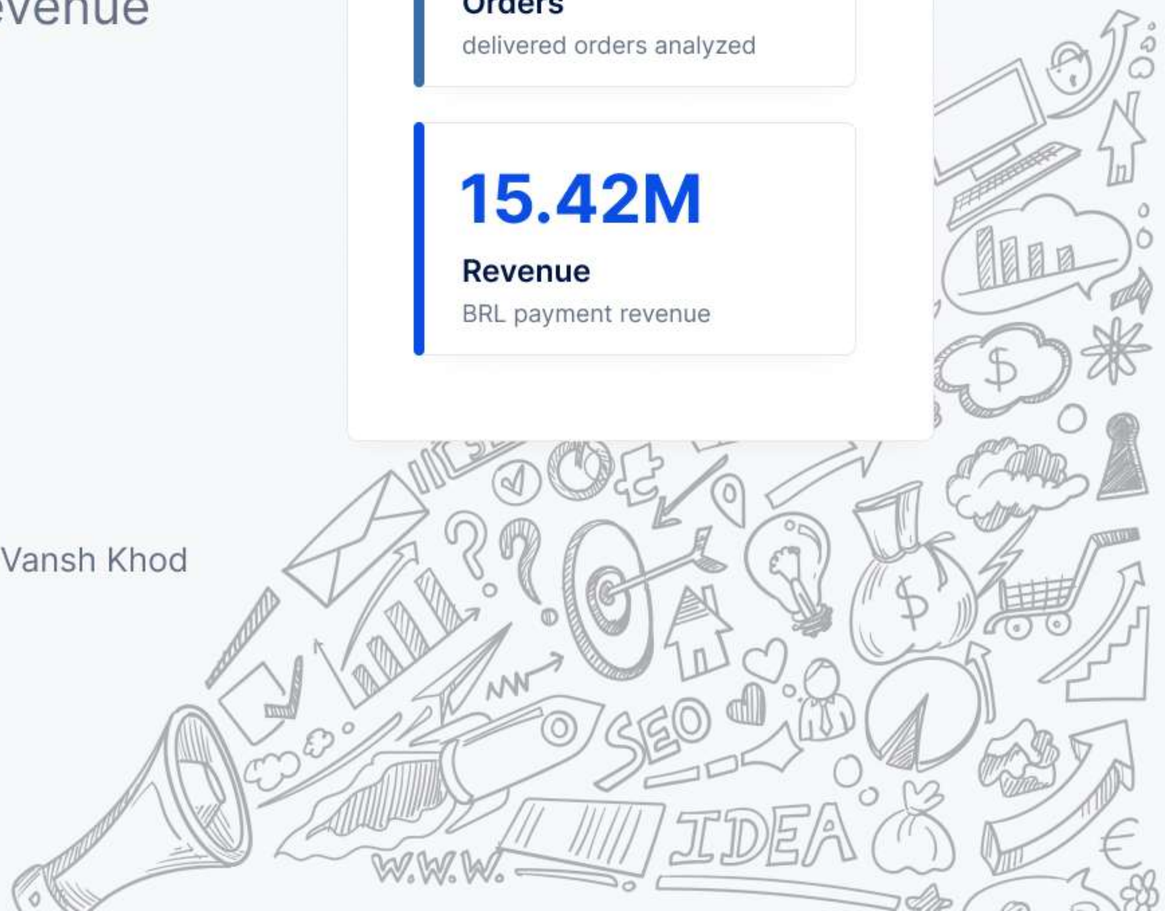
**Marketplace scale**

**96.5K**
**Orders**
delivered orders analyzed

**15.42M**
**Revenue**
BRL payment revenue



# Context & Problem Statement

Olist connects small and medium sellers to customers across all 27 Brazilian states.

CORE BUSINESS QUESTION

**Which operational levers most influence customer satisfaction and revenue, and where should Olist invest first?**

## The challenge

- Review outcomes span 1 to 5 with a polarized distribution
- Delivery delays vary meaningfully by region

**VP Ops**
**Stakeholder**
Marketplace operations decision owner

**27**
**States**
Brazilian states represented

**3.0%**
**Repeat Rate**
repeat purchase gap

**1-5**
**Review Range**
primary satisfaction metric

**Decision supported: allocate improvement budgets across logistics, seller/category quality, payment strategy, and retention with quantified evidence.**

# Data Engineering

A reproducible Python pipeline turns 9 relational CSVs into one analysis-ready order-item table and Tableau exports.

### Raw source

Olist Brazilian E-Commerce
9 relational CSVs
~99K orders
Sep 2016 - Oct 2018

### ETL & QA

Datetime parsing
Payment aggregation
Review deduplication
Category translation fixes
Geolocation aggregation

### Outputs

110,197 rows x 48 cols
96,478 order KPI rows
5 Tableau KPI exports
Interactive dashboard

**Join strategy**

Orders backbone → customers → items → products → sellers → payments aggregated to order level → latest review per order. Geolocation is aggregated before joining to avoid row explosion.

OLIST MARKETPLACE ANALYTICS

# KPI Framework

Seven KPIs connect operational performance to satisfaction and revenue efficiency.

**BRL 15.42M**
**Total Revenue**
sum of order payments

**BRL 159.86**
**AOV**
average order value

**4.16 / 5**
**CSAT**
average review score

**93.2%**
**On-Time Delivery**
delivered on/before ETA

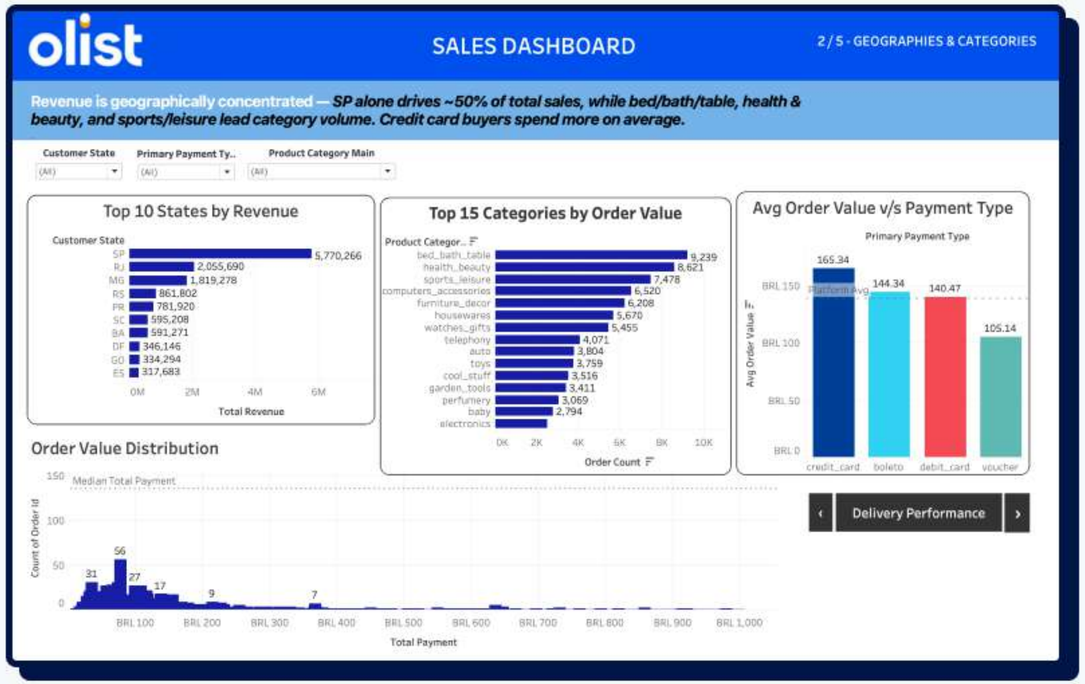
**3.0%**
**Repeat Purchase**
customers with 2+ orders

**16.6%**
**Freight Ratio**
platform freight / item value

**Why these matter: each KPI ties directly to one of two outcomes leadership can act on - customer satisfaction or revenue efficiency.**

# Key EDA Insights

Revenue, geography, category mix, and review behavior reveal where operational pressure is concentrated.




- Revenue nearly doubled from Jan to Nov 2017, with a Black Friday spike.
- SP, RJ, and MG dominate revenue, creating Southeast concentration risk.
- High-revenue categories such as bed/bath/table, computers, and furniture show satisfaction leakage.
- Review scores are bimodal: most customers give 5 stars, but 1-star failures are a large minority.

**Operational reading: growth is real, but customer experience is uneven across state, category, and logistics contexts.**

# Advanced Analysis

Formal tests validate that delivery timeliness is the strongest satisfaction lever.

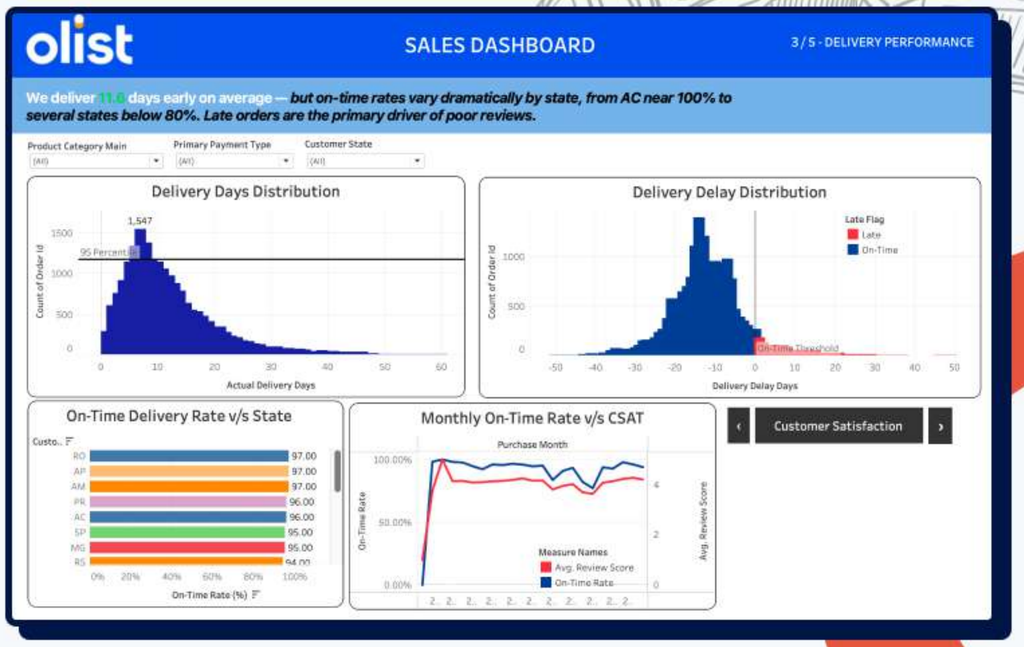
## Statistical proof

Welch t-test: late orders have much lower review scores

Chi-square: review score and on-time status are strongly associated

OLS regression: is_late is the largest negative coefficient

ANOVA: payment type significantly affects order value




**Late delivery effect**

On-time orders: 4.29 / 5

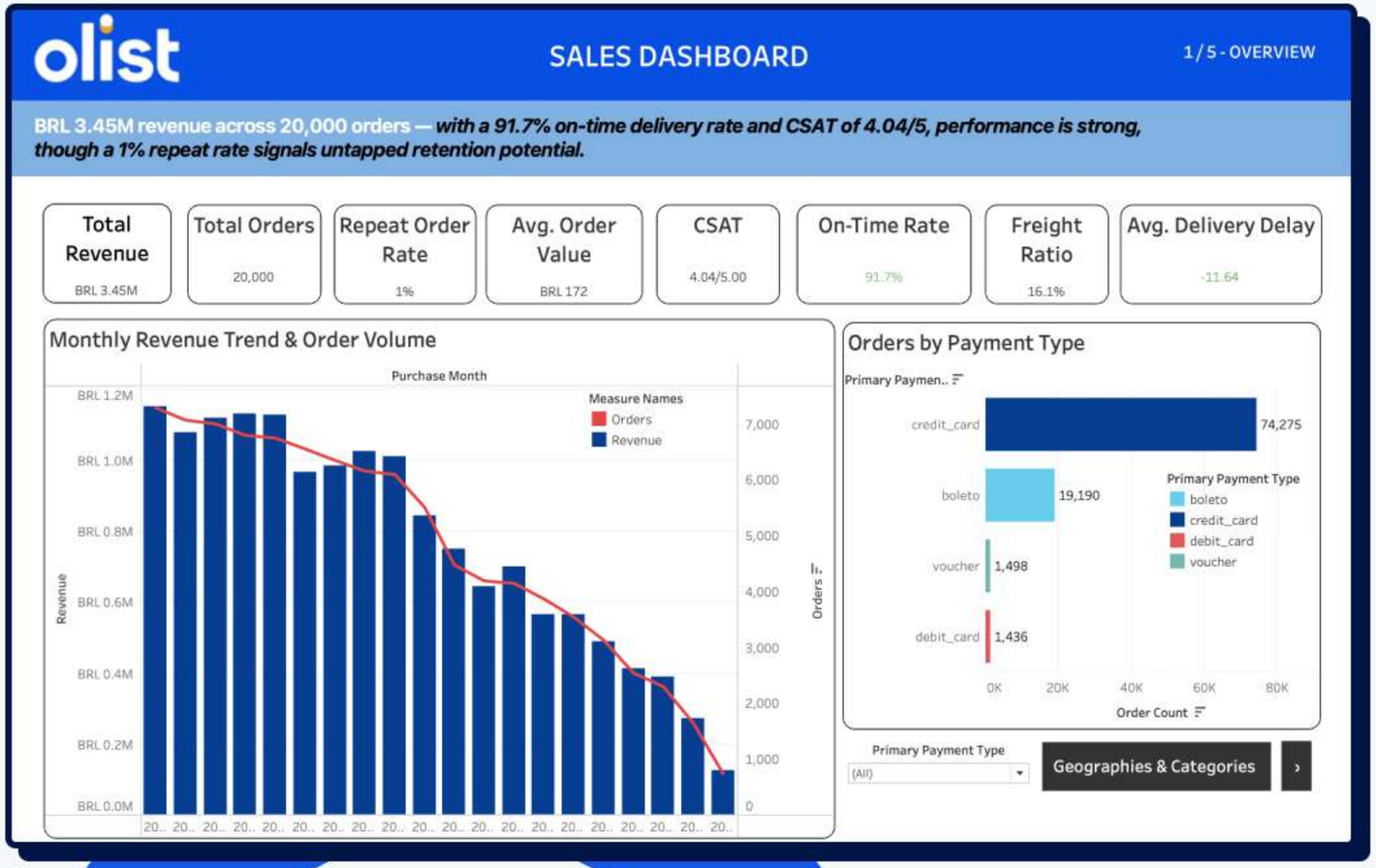
Late orders: 2.27 / 5

**Cohen's d = 1.467; Cramer's V = 0.418**

The binary shock of missing the promised date matters more than small changes in raw delivery speed.

# Dashboard Overview

The Tableau workbook turns the analysis into an executive and operational decision surface.




## How to read it

- Executive scorecard for revenue, AOV, CSAT, and on-time delivery
- Operational drilldowns by state, category, payment type, and review band
- Monthly trends expose seasonality and delivery quality shifts
- Filters support leadership questions instead of static reporting

public.tableau.com/app/profile/hemanth.tenneti/viz/SectionB_Group10_OlistCommerce/Overview

# Top Insights

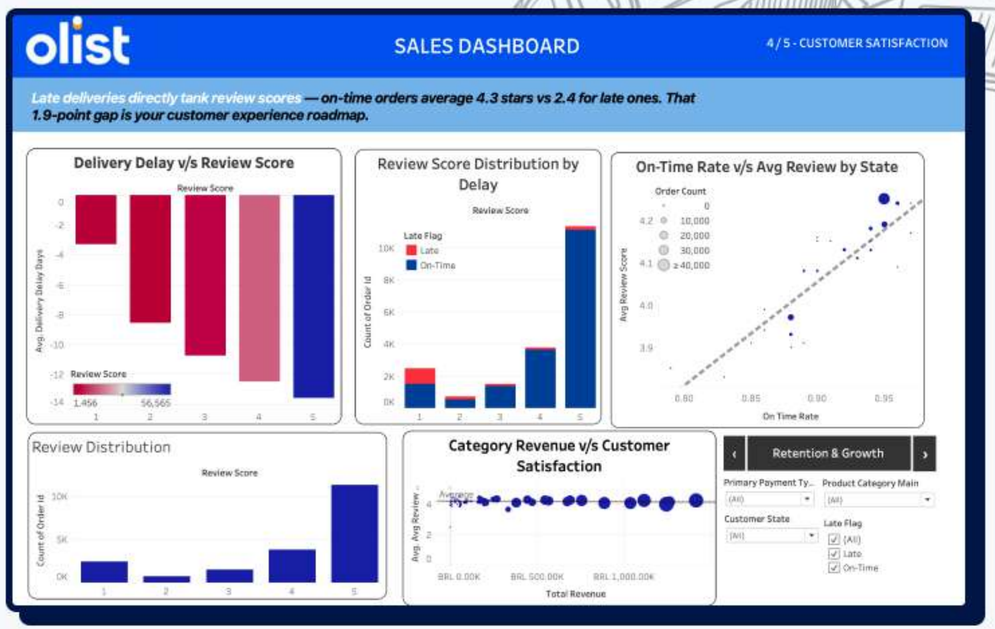
Decision language: what leadership should take away from the evidence.

1. **Late delivery is a satisfaction killer**
   On-time orders average 4.29; late orders average 2.27.

2. **97% of customers never come back**
   Retention is the largest untapped growth lever.

3. **Promise adherence beats raw speed**
   Meeting the ETA matters more than shaving every delivery by a day.

4. **Bulky categories leak satisfaction**
   Furniture, office furniture, and computers need better logistics controls.




**The business problem is not only growth. It is controlled growth with reliable delivery and repeatable customer trust.**

# Recommendations

Five focused actions translate the analysis into operational priorities.

### 1 Set 95% on-time SLA targets

Tie carrier and seller performance to promised delivery adherence.

### 2 Launch 7-day post-purchase nurture

Convert first-time buyers with tracking, review prompts, and recommendations.

### 3 Build regional logistics partnerships

Reduce North/Northeast freight burden through carrier rates and forward hubs.

### 4 Create a bulky-category logistics tier

Protect furniture, appliances, and computers with premium handling protocols.

### 5 Use payment campaigns to lift AOV

Promote installments and bundle offers for high-ticket or boleto segments.

# Impact & Value

Expected outcomes should be presented as directional estimates, then validated through pilots.

**+0.3 to +0.5**

**CSAT Uplift**

expected points from delivery focus

**+15-20%**

**Retention Revenue**

from repeat purchase uplift

**15-20%**

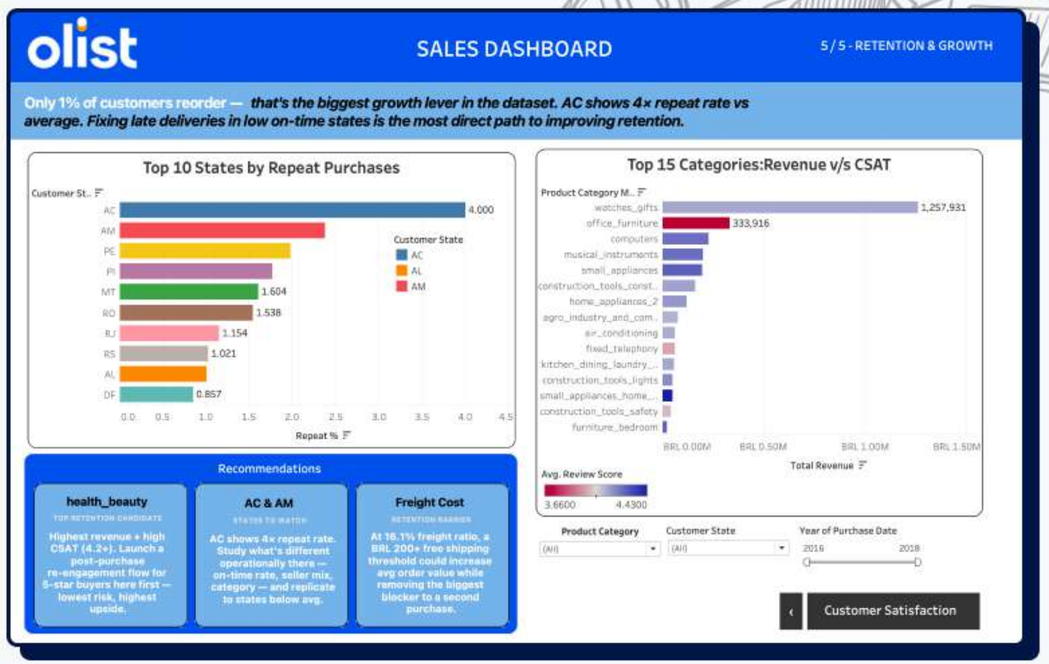
**Freight Reduction**

North/Northeast target

**+8-12%**

**AOV Growth**

targeted payment segments




**Priority ranking: On-time SLA > Post-purchase nurture > Regional logistics > Bulky-category tier > Payment partnerships**

# Limitations

The analysis is decision-ready, but not a causal proof of every proposed intervention.

### Data window

Dataset covers 2016-2018; current marketplace operations may differ.

### Missing demographics

No age, income, gender, or household attributes are available.

### Seller economics

Margins, commissions, and seller cost structures are not included.

### Correlation limits

Statistical tests show association, not full causal identification.

### Text sparsity

Review comments are sparse and Portuguese-language NLP is future work.

# Next Steps

Move from retrospective analytics to operational experimentation and monitoring.

1

### Predictive churn model

Gradient boosting for customer-level retention scoring

2

### Seller performance index

Automated seller rating tied to delivery and review metrics

3

### A/B testing framework

Test delivery estimate accuracy against CSAT lift

4

### Portuguese review NLP

Extract qualitative satisfaction drivers from sparse text

5

### Real-time KPI dashboard

Move from static Tableau exports to live monitoring

**Dashboards do not get marks. Decisions get marks. The decision is clear: fix delivery promise adherence, then convert first-time buyers into repeat customers.**